Interference in Thin Films and Transfer Matrix Formalism

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Abstract

Thin-film interference is a ubiquitous optical phenomenon responsible for the colourful & iridescent reflections that we see in oil films on water and soap bubbles. It also has applications in the construction of special optical devices. This paper delves into the fundamental principles of thin-film interference, derives mathematical conditions for constructive and destructive interference, and discusses its various real-world applications. Furthermore, it establishes the concept of transfer matrices, and explores it usefulness as a modality to understand complicated multilayer systems.

1. Introduction

Although there does not exist a precise definition, thin films can be considered as layers of material with thicknesses ranging from a few nanometres to several micrometres. When light interacts with these films, it undergoes multiple reflections and transmissions at the interfaces, leading to interference effects. This phenomenon, known as thin-film interference, results in striking optical patterns that are widely observed in nature and utilized in various technological applications.

Interference in thin films occurs due to the phase differences between light waves reflecting off the upper and lower surfaces of the film. Depending on the thickness of the film, the wavelength of light, and the refractive indices of the materials involved, these reflections can interfere constructively or destructively, producing a spectrum of colours or enhancing or suppressing specific wavelengths.

This effect is responsible for the vibrant colours seen in soap bubbles (See figure 1) and oil slicks, as well as for engineered coatings that enhance or reduce reflectivity in optical devices. Applications range from anti-reflective coatings on eyeglasses and camera lenses to high-reflectivity mirrors in telescopes and lasers. Thin-film interference also plays a crucial role in precision optical instruments such as Fabry-Pérot interferometers, which are used in spectroscopy and metrology.

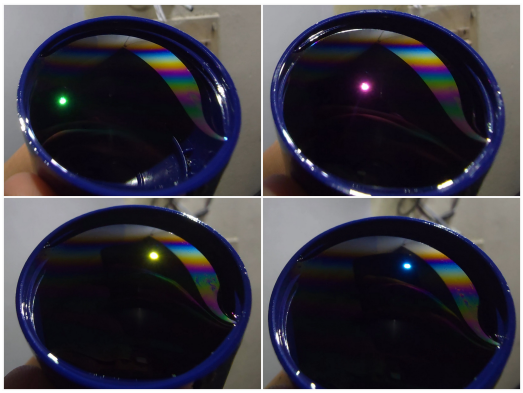
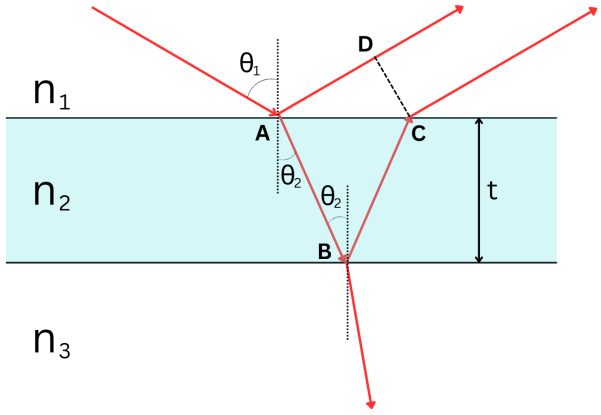
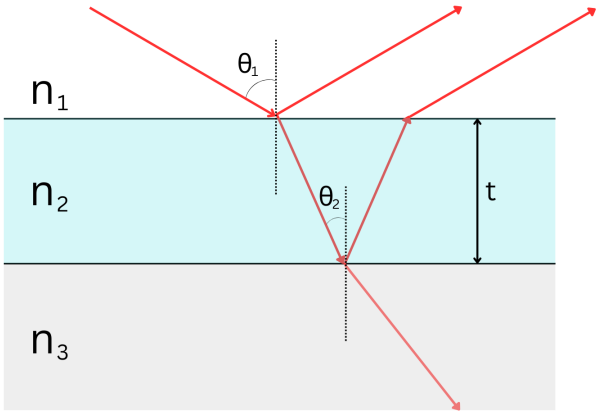


Figure 1: Thin-film interference in a soap bubble. A phone flashlight was moved across, illuminating different regions and revealing distinct colours corresponding to varying thicknesses.

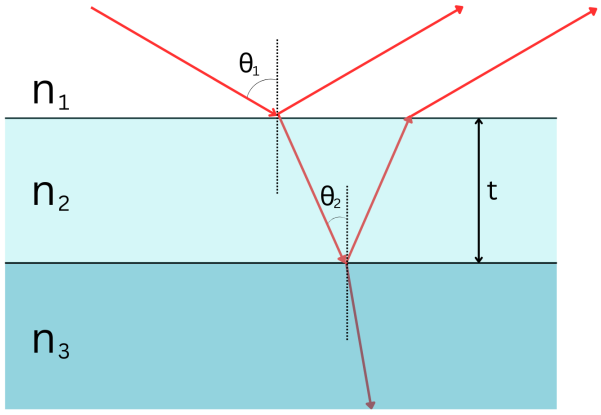
2. Interference in thin films



(a)



(b)



(c)

Figure 2: (a) schematic diagram showing the optical path difference between two reflected rays,
(b) thin film interference when the film lies over a material with less refractive index,
(c) thin film interference when the film lies over a material with more refractive index.

2.1. Optical Path Difference

With reference to figure 2(a), the optical path difference OPD is given by:

$$OPD = n_2(AB + BC) - n_1 AD$$

The optical path along $A \to B \to C$ is:

$$2AB = \frac{t}{\cos\theta_2}$$

The optical path along AD is:

$$AD = (2t\tan\theta_2)\sin\theta_1$$

Therefore, we get the OPD to be:

$$\begin{aligned} OPD &= 2tn_2\left[\frac{1}{\cos\theta_2} - \tan\theta_2\sin\theta_2\right] \\ &= 2tn_2\left[\frac{1-\sin^2\theta_2}{\cos\theta_2}\right] = 2tn_2\cos\theta_2 \end{aligned} \tag{1}$$

2.2. Thin film in air ($n_1 < n_2 > n_3$)

Refer to figure 2(b). A thin film of refractive index n_2 and thickness t is placed between two media: air (n_1) and another medium (n_3), where $n_1 < n_2 > n_3$. A light wave is incident from air at an angle θ_1. We notice two things. Reflection at the first interface ($n_1 \to n_2$) undergoes a π phase shift (180°) since $n_2 > n_1$. Also, reflection at the second interface ($n_2 \to n_3$) undergoes no phase shift if $n_2 > n_3$.

The optical path difference (OPD) due to the extra travel inside the film is given by:

$$\text{OPD} = 2tn_2\cos\theta_2$$

where θ_2 is the refracted angle inside the film.

- For constructive interference (bright fringes): $2tn_2\cos\theta_2 = \left(m + \frac{1}{2}\right)\lambda$.
- For destructive interference (dark fringes): $2tn_2\cos\theta_2 = m\lambda$.

Here m is an integer ($m = 0, 1, 2, \ldots$). The $\frac{1}{2}\lambda$ term accounts for the π phase shift.

2.3. Thin film on a denser medium ($n_1 > n_2 < n_3$)

Refer to figure 2(c). A thin film of refractive index n_2 is placed between two media, where $n_1 > n_2 < n_3$. A light wave is incident from n_1. Considering phase inversions, reflection at the first interface ($n_1 \to n_2$) undergoes no phase shift as $n_1 > n_2$, reflection at the second interface ($n_2 \to n_3$) undergoes a π phase shift if $n_3 > n_2$. The optical path difference remains:

$$\text{OPD} = 2tn_2 \cos\theta_2$$

- For constructive interference (bright fringes): $2tn_2 \cos\theta_2 = m\lambda$.
- For destructive interference (dark fringes): $2tn_2 \cos\theta_2 = \left(m + \frac{1}{2}\right)\lambda$.

3. Everyday examples of thin film interference

3.1. Soap bubbles

The film comprising a soap bubble consists of a mixture of water and lipids (soap molecules). The entire film, which can be thought to have an effective refractive index close to that of water, corresponds to the case of figure 2 with $n_1 = n_3$. We can explain the different colours observed as follows: since the refractive index of water is different for different wavelengths of light, different colours would exhibit constructive and destructive interference at different thicknesses. Since the flow inside the soap film is usually turbulent and the thickness is non-uniform, we observe regions with different colours and they keep changing with time.

3.2. Butterfly wings and Peacock feathers

Blue morpho butterflies, known for their brilliant metallic blue wings, owe their colour to the structural features of their wings. Their wings consist of a colourless translucent membrane covered by a layer of scales with ridges. These ridges and scales lead to multiple reflecting surfaces. We have only discussed single film interference till here, but the concepts developed can be extended to multiple layers using matrix formulation. The spacing of ridges on the scales is at just the right distance to cause constructive interference of light in the blue portion of the colour spectrum when the light strikes—and is reflected—from the wing (at a certain angle). Peacock feathers too, use the same principle to get their astonishing colours[4].

Before we move on to other sections, an important point that should be discussed is the possibility of occurrence of these phenomena with thick films. One can, of course, ponder upon why these phenomena are not observed with thick films in daily life - because eventually, a phase difference between the two reflected rays is the ultimate requirement. However, one should not overlook the need for coherence between the interfering rays.
This is what thick films lack. Due to the small coherence length of thermal sources we use in our daily lives, the rays reflecting of the two surfaces of a thick film will no longer be coherent. If the two interfering rays are not coherent, the phase difference would change rapidly and one would not observe only the time averaged intensities. This is why we observe this phenomenon with only thin films - because we are surrounded by thermal sources.

4. Stratified media

A medium composed of homogeneous layers of material, each of which differs in their electromagnetic properties, is said to be stratified in nature. Impedance mismatch between the constituent layers is a characteristic property of stratified media. Multiple refraction and transmission effects are seen to occur at the interfaces within such media, each of which gives rise to new wave

components with varying phase shifts. Since the type and extent of interference are inherently dependent upon the phases and amplitudes of the superimposing waves, the phase information as well as the reflection and transmission probabilities provided by each of these interfaces become important.

5. Introduction to transfer matrices

A mathematically rigorous approach that can be employed to study the propagation of electromagnetic radiation in stratified media is the method of transfer matrices.

5.1. Definition

A transfer matrix M for a particular interface is defined as;

$$M = \begin{bmatrix} \alpha & \beta \\ \beta^* & \alpha^* \end{bmatrix}$$

where, α & β are known as Bogoliubov coefficients such that $|\alpha|^2$ - $|\beta|^2 = 1$

In terms of the reflection and transmission amplitudes associated with the interface under consideration, we have;

$$\alpha = 1/t \;\&\; \beta = r/t$$
$$\Rightarrow M = \begin{bmatrix} 1/t & r/t \\ r^*/t^* & 1/t^* \end{bmatrix} \tag{2}$$

The Bogoliubov coefficients can be parametrized using real numbers $\{\theta, \phi_a, \phi_b\}$ in the following manner[1];

$$\alpha = e^{i\phi_a}\cosh(\theta) \;\&\; \beta = e^{i\phi_b}\sinh(\theta)$$
$$\Rightarrow t = e^{-i\phi_a}\,\mathrm{sech}(\theta) \;\&\; r = e^{-i(\phi_a-\phi_b)}\tanh(\theta) \tag{3}$$

Since the reflection and transmission probabilities of an interface are defined as $\mathrm{R} = |r|^2$ and $\mathrm{T} = |t|^2$ respectively. we have;

$$\mathrm{T} = \frac{1}{|\alpha|^2} = \mathrm{sech}^2\theta \tag{4}$$

$$\mathrm{R} = \frac{|\beta|^2}{|\alpha|^2} = \tanh^2\theta \tag{5}$$

5.2. Properties

- Each interface in a stratified medium has a characteristic transfer matrix
- When the ith interface in a stratified medium is moved by a distance a_i, the transfer matrix of the interface gets modified in the following manner[2];

$$M_i \rightarrow \tilde{M}_i = \begin{bmatrix} \alpha_i & \beta_i e^{i2ka_i} \\ \beta_i^* e^{-i2ka_i} & \alpha_i^* \end{bmatrix}$$

  The α Bogoliubov coefficient and, therefore, the transmission amplitude t_i remains invariant under a shift in the physical location of the barrier. The β Bogoliubov coefficient and, therefore, the reflection amplitude r_i picks up a shift-dependent e^{i2ka_i} phase.

- The compound transfer matrix for a stratified media with n non-overlapping interfaces in given by[3];

$$M_{1,2\ldots,n} = M_1 M_2 \ldots M_n \tag{6}$$

6. Deep dive into two-interface systems

Consider a medium that has two interfaces. Here, the compound transfer matrix will be given by;

$$M_{12} = M_1 M_2 = \begin{bmatrix} \alpha_1\alpha_2 + \beta_1\beta_2^* & \alpha_1\beta_2 + \beta_1\alpha_2^* \\ \alpha_1^*\beta_2^* + \beta_1^*\alpha_2 & \alpha_1^*\alpha_2^* + \beta_1^*\beta_2 \end{bmatrix}$$

[1] Refer to section A.1.1

[2] Refer to section A.1.2

[3] Refer to section 7

$$\text{where,}\alpha_{12} = \alpha_1\alpha_2 + \beta_1\beta_2^*$$
$$\beta_{12} = \alpha_1\beta_2 + \beta_1\alpha_2^*$$

The upper and lower bounds of α_{12} & β_{12} in the parametric form can be defined in the following manner[4];

$$|\alpha_{12}| \leq \cosh\{\cosh^{-1}|\alpha_1| + \cosh^{-1}|\alpha_2|\}$$
$$|\beta_{12}| \leq \sinh\{\sinh^{-1}|\beta_1| + \sinh^{-1}|\beta_2|\}$$

$$|\alpha_{12}| \geq \cosh\{\cosh^{-1}|\alpha_1| - \cosh^{-1}|\alpha_2|\}$$
$$|\beta_{12}| \geq \sinh\{\sinh^{-1}|\beta_1| - \sinh^{-1}|\beta_2|\}$$

The upper bounds on the Bogoliubov coefficients of the two-interface system lead to a lower bound on transmission probability (T_{12}) and an upper bound on the reflection probability (R_{12}) of the system. Therefore, we have;

$$T_{12} \geq \text{sech}^2\left(\text{sech}^{-1}\sqrt{T_1} + \text{sech}^{-1}\sqrt{T_2}\right)$$
$$R_{12} \leq \tanh^2\left(\tanh^{-1}\sqrt{R_1} + \tanh^{-1}\sqrt{R_2}\right)$$

In a similar fashion, the lower bounds on the Bogoliubov coefficients are seen to translate into an upper bound on transmission probability (T_{12}) and a lower bound on the reflection probability (R_{12}) of the system. That is;

$$T_{12} \leq \text{sech}^2\left(\text{sech}^{-1}\sqrt{T_1} - \text{sech}^{-1}\sqrt{T_2}\right)$$
$$R_{12} \geq \tanh^2\left(\tanh^{-1}\sqrt{R_1} - \tanh^{-1}\sqrt{R_2}\right)$$

Manipulation of the above hyperbolic and the inverse hyperbolic functions using elementary identities leaves us with the following relations[5];

$$T_{12} \geq \frac{T_1T_2}{\left(1-\sqrt{1-T_1}\sqrt{1-T_2}\right)^2}$$
$$T_{12} \leq \frac{T_1T_2}{\left(1+\sqrt{1-T_1}\sqrt{1-T_2}\right)^2}$$
$$R_{12} \geq \left\{\frac{\sqrt{R_1}-\sqrt{R_2}}{1-\sqrt{R_1}\sqrt{R_2}}\right\}^2$$

[4]Refer to section A.2.1

[5]Refer to section A.2.2

$$R_{12} \leq \left\{\frac{\sqrt{R_1}+\sqrt{R_2}}{1+\sqrt{R_1}\sqrt{R_2}}\right\}^2$$

Thus, the reflection and transmission probabilities of a two-interface system are seen to oscillate between non-trivial upper and lower bounds.

Since $\{R_{12}, T_{12}\} \epsilon [0,1]$, we get;

$$\frac{\sqrt{R_1}+\sqrt{R_2}}{1+\sqrt{R_1}\sqrt{R_2}} \leq 1$$
$$\frac{T_1T_2}{\{1+\sqrt{1-T_1}\sqrt{1-T_2}\}} \leq 1$$

Consider a situation in which the waves of a specific wavelength that have been reflected off the interfaces in our system of study destructively interfere with each other. As principle of energy conservation holds true in an ideal and lossless system, the effect of this interference is the redirection of these wavelengths into the transmitted component. As $R \to 0$, $T \to 1$, resulting in highly enhanced transmission effects for specific wavelengths. This is known as transmission resonance.

To get an exact transmission resonance ($T_{12} = 1$) for our two-interface system, it is necessary that;

$$\frac{T_1T_2}{\{1+\sqrt{1-T_1}\sqrt{1-T_2}\}} = 1 \quad (7)$$
$$\Rightarrow T_1 = T_2 \quad (8)$$

Thus, an exact transmission resonance from two non-overlapping interfaces is possible only if the the two interfaces have equal transmission probabilities!

7. Transfer matrix in multilayer systems

The necessity of transfer matrix formalism is clearly evident when we extend it to multilayered systems, in which finding reflection and transmission coefficients of each layer becomes very convoluted very quickly, as we will have to consider reflection and transmission of every single layer to determine the coefficients at each layer.

The multilayer structure is shown in Figure 3 has N

layers and $N+1$ interfaces. The superscripts + and − indicate right and left going waves. The amplitudes of the field at the left- and right-hand side of an interface are related by:

$$\begin{pmatrix} E_{m-1}^{+} \\ E_{m-1}^{-} \end{pmatrix} = \mathbf{I}_{m-1,m} \begin{pmatrix} E_{m}^{'+} \\ E_{m}^{'-} \end{pmatrix},$$

$$\mathbf{I}_{m-1,m} = \frac{1}{t_{m-1,m}} \begin{pmatrix} 1 & r_{m-1,m} \\ r_{m-1,m} & 1 \end{pmatrix} \tag{9}$$

where, $\mathbf{I}_{m-1,m}$ is called the refraction or transmission matrix of the interface and $t_{m-1,m}$ and $r_{m-1,m}$ are the transmission and reflection Fresnel coefficients, respectively. The field amplitudes at the left- and right-hand side of the mth layer are related by:

$$\begin{pmatrix} E_{m}^{'+} \\ E_{m}^{'-} \end{pmatrix} = \mathbf{P}_m \begin{pmatrix} E_{m}^{+} \\ E_{m}^{-} \end{pmatrix}, \ \mathbf{P}_m = \begin{pmatrix} e^{i\delta_m} & 0 \\ 0 & e^{-i\delta_m} \end{pmatrix} \tag{10}$$

where $\mathbf{P}_m$ is called the propagation matrix of the layer.

$$\delta_m = 2\pi\sigma n_m d_m \tag{11}$$

δ_m is the phase shift due to wave passing through the film m, σ is the wave number $(1/\lambda_0)$, n_m and d_m are the refractive index and the thickness respectively. The derivation of these matrices have been skipped in this paper considering their length. The section in [3] has a complete derivation of these matrices.

Now, if we consider just the layer m along with its contact interfaces, which is like a single layer; thus we have;

$$\begin{pmatrix} E_{m-1}^{+} \\ E_{m-1}^{-} \end{pmatrix} = \mathbf{I}_{m-1,m}\mathbf{P}_m\mathbf{I}_{m,m+1} \begin{pmatrix} E_{m+1}^{'+} \\ E_{m+1}^{'-} \end{pmatrix} \tag{12}$$

The iterative application of these transformations for the N layers and $N+1$ interfaces will give the Transfer Matrix $\mathbf{M}$ of the entire system.

$$\begin{pmatrix} E_{0}^{+} \\ E_{0}^{-} \end{pmatrix} = \mathbf{I}_{0,1}\mathbf{P}_1\mathbf{I}_{1,2}\mathbf{P}_2\mathbf{I}_{2,3}\cdots\mathbf{I}_{N-1,N}\mathbf{P}_N\mathbf{I}_{N,N+1} \begin{pmatrix} E_{N+1}^{+} \\ E_{N+1}^{-} \end{pmatrix}$$

$$\mathbf{M}_{0,N+1} = \mathbf{M} = \mathbf{I}_{0,1}\mathbf{P}_1\mathbf{I}_{1,2}\mathbf{P}_2\mathbf{I}_{2,3}\cdots\mathbf{I}_{N-1,N}\mathbf{P}_N\mathbf{I}_{N,N+1}$$

It is possible to write $\mathbf{I}_{m-1,m}$ as a product of two matrices. i.e $\mathbf{I}_{m-1,m} = \mathbf{D}_{m-1}^{-1}\mathbf{D}_m$ where the matrix $\mathbf{D}_i$ is the Dynamical Matrix representing i^{th} side of an interface.

$$\mathbf{D_i} = \begin{pmatrix} 1 & 1 \\ n_i\cos\theta_i & -n_i\cos\theta_i \end{pmatrix} \tag{13}$$

for an s-polarized wave, where n_i is the index of refraction of the i_{th} layer and θ_i is the ray angle i^{th} layer. Now writing the above in terms of $\mathbf{D}$ we get;

$$\begin{pmatrix} E_{0}^{+} \\ E_{0}^{-} \end{pmatrix} = \mathbf{D}_0^{-1}\left[\prod_{m=1}^{N}\mathbf{D}_m\mathbf{P}_m\mathbf{D}_m^{-1}\right]\mathbf{D}_{N+1} \begin{pmatrix} E_{N+1}^{+} \\ E_{N+1}^{-} \end{pmatrix} \tag{14}$$

Defining $\mathbf{M}_m = \mathbf{D}_m\mathbf{P}_m\mathbf{D}_m^{-1}$ and $\mathbf{M}_0 = \mathbf{D}_0^{-1}$, $\mathbf{M}_{N+1} = \mathbf{D}_{N+1}$ where $\mathbf{M}_i$ represents the transfer matrix at the ith layer. We can now rewrite the equation of $\mathbf{M}$ in a familiar form as seen in previous section.

$$\begin{pmatrix} E_{0}^{+} \\ E_{0}^{-} \end{pmatrix} = \mathbf{M}_0\mathbf{M}_1\cdots\mathbf{M}_N\mathbf{M}_{N+1} \begin{pmatrix} E_{N+1}^{+} \\ E_{N+1}^{-} \end{pmatrix} \tag{15}$$

$$\mathbf{M} = \mathbf{M}_0\mathbf{M}_1\cdots\mathbf{M}_N\mathbf{M}_{N+1} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \tag{16}$$

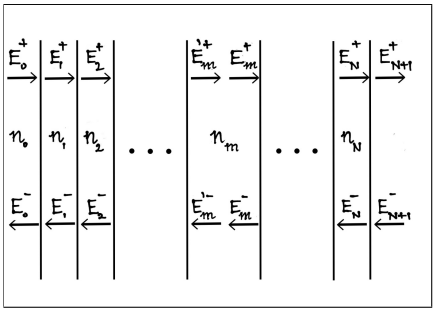


Figure 3: Multilayer system composed of N layers. The prime is used for waves at the right hand side of an interface.

8. Modelling partially coherent and incoherent interference in multi-layer systems

Partial coherence and incoherence can be introduced by adding a random phase to the phase shift in a required layer. Thus, we can rewrite the δ_m equation (11) as

$$\delta_m = 2\pi\sigma n_m d_m + \beta_m \Theta_m$$

where β_m takes values between 0 and π, and Θ_m is randomly generated number between -1 and 1 (uniformly distributed). The final transmittance is obtained by averaging the calculated transmittances with different sequences of random numbers. We can, because of how we formulated the phase, have different degrees of incoherence in each layer. It has been found that total incoherence is best represented by introducing a random phase with $\beta_m = \pi$ in this methodology, while for partial coherence cases, $0 < \beta < \pi$.

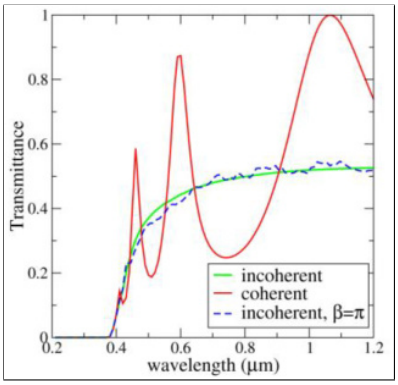


Figure 4: Transmittance vs wavelength for a crystalline Si film of 150 nm thickness

Figure 4 shows a comparison between the squares of the amplitude results (green line) with our random phase method (dashed line) for the case of incoherent limit ($\beta = \pi$). We can see a really good fit. Similar comparisons can be done with partial coherent limit [1], which also agree with our formalism.

It is also possible to follow an approach using modified Fresnel coefficients and the square of their amplitudes as done in [2] to work in the incoherent limit but our approach is better as the addition of a random phase can be physically interpreted as a means to simulate the loss of coherence caused by defects present in the layer.

In summary, we have developed a method for calculating the optical response of multilayer systems, which can deal with coherent, partially coherent, and incoherent interference on equal footing.

9. Applications of thin film interference and matrix formalism

9.1. Anti-reflective coatings

Anti-reflective coatings minimize the reflection of one or more wavelengths and are typically used on the surface of lenses so that less light is lost. They are also used on smartphone screens to make the screen clearly visible in daylight. [6]

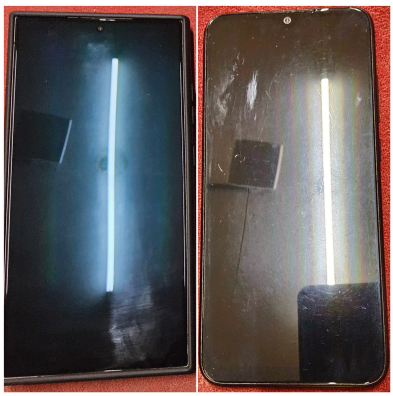


Figure 5: smartphone with anti-reflective coating (left) vs. without anti-reflective coating (right)

This also reduces stray reflected light in sensitive optical instruments. For an air glass interface and normal incidence, the reflectivity is around 4%. This can be reduced by coating an intermediate layer between the air glass interface. We consider the simplest of these coatings - the configuration being similar to figure 2, with $n_1 = n_{air}$, $n_2 = n_{coating}$, $n_3 = n_{glass}$.

We label the air-coating interface is surface 1 and

coating-glass interface as surface 2. From section 6 equation (8), for maximum transmission (or minimum reflection), the transmission probabilities for both surfaces should be equal.

$$\therefore T_1 = T_2 \Rightarrow |t_1|^2 = |t_2|^2$$

We consider normal incidence. t_1 and t_2 are Fresnel transmission coefficients. Putting in their values for normal incidence,

$$\left(\frac{2n_1}{n_1+n_2}\right)^2 = \left(\frac{2n_2}{n_2+n_3}\right)^2$$
$$\Rightarrow 2n_1n_2 + 2n_1n_3 = 2n_2n_1 + 2n_2^2$$
$$\therefore n_2 = \sqrt{n_1n_3}$$

For a typical glass lens, $n_1 = n_{air} = 1$, $n_3 = 1.52$ which gives us $n_2 \approx 1.23$. Materials with such low refractive index are not found, so the reflectivity does not really go to zero. However, using multiple layers, and also setting the thickness such that the reflecting waves destructively interfere, the reflectivity can be decreased even more.

9.2. Bragg reflectors

Bragg reflectors are special mirrors which display very high reflectivity for certain wavelengths. They consist of multiple coating layers, with alternate indices of refraction, depending on the principal wavelength to be reflected. The most frequently used design is that of a quarter-wave mirror, where each optical layer thickness corresponding to one quarter of the wavelength for which the mirror is designed. Consider such a Bragg reflector with the same configuration as shown in figure 3, but $n_1 = n_3 = n_5... = n_a$ and $n_2 = n_4 = n_6... = n_b$. The refractive indices of initial medium and final medium are n_0, n_g respectively. Each layer has thickness d. Therefore from equation (13) we have, for normal incidence:

$$D_m = \begin{pmatrix} 1 & 1 \\ n_m & -n_m \end{pmatrix} \tag{17}$$

where $n_m = n_a$ if $m = odd$ and $n_m = n_b$ if $m = even$. Also, using equation (11) taking $\phi_m = 0^\circ$, we have

$$\delta_m = 2\pi\sigma n_m d = \pi/2 \tag{18}$$

where we used the fact that typical Bragg reflectors are quarter wave stacks, i.e. $n_m d = \lambda_0/4 = 1/4\sigma$. This allows the phase difference $\delta_m = \pi/2$ for all m. This makes $e^{i\delta_m} = -i$ and $e^{-i\delta_m} = i$ for all m. Using equations (18) and (10) we can write P_m, and then using the definition of $M_m = D_m P_m D_m^{-1}$, we have

$$M_m = \begin{pmatrix} 1 & 1 \\ n_m & -n_m \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \frac{1}{2n_m} \begin{pmatrix} n_m & 1 \\ n_m & -1 \end{pmatrix}$$

On solving,

$$M_m = \begin{pmatrix} 0 & i/n_m \\ in_m & 0 \end{pmatrix} \text{ for m=1,2,3,...N} \tag{19}$$

Also, by definition

$$M_0 = \frac{1}{2n_0}\begin{pmatrix} n_0 & 1 \\ n_0 & -1 \end{pmatrix}, M_{N+1} = \begin{pmatrix} 1 & 1 \\ n_g & -n_g \end{pmatrix} \tag{20}$$

Using equations (16), (19), (20), and the fact that $n_m = n_a$ if $m = odd$ and $n_m = n_b$ if $m = even$, we can obtain the transfer matrix of the entire system. By definition, using equation (2), Reflectivity

$$R = |r|^2 = |\frac{M_{21}}{M_{11}}|^2$$

On calculating the Final transfer matrix and using the above equation, we get

$$R = \left(\frac{1-(n_g/n_0)(n_a/n_b)^{2N}}{1+(n_g/n_0)(n_a/n_b)^{2N}}\right)^2$$

It is clear that irrespective whether $n_a > n_b$ or not, the reflectivity exponentially approaches 1, as N is increased. Therefore by tuning the thicknesses and refractive indices of the layers, we can design efficient reflectors for different wavelengths.

9.3. Optical filters

Optical filters use multiple interfaces to either block certain wavelengths or transmit certain wavelengths. Filters that transmit a certain band of frequencies/wavelengths are called bandpass filters while those which block a certain range are called notch filters. These filters consist of stacks of media with different refractive indices. By controlling the thickness of each film and the refractive index, filters can be tuned such that certain transmitted wavelengths destructively interfere with each other while others constructively interfere. Transfer matrices can be used to find the optimum refractive indices and thickness of each layer.

Apart from these applications, thin films are also used in OLED, LCD displays to enhance colour reproduction, in photolithography[7] as well, to design nanoparticles, and also in biosensors.

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Appendix

A.1 Introduction to transfer matrices

A.1.1 Parametrization of Bogoliubov coefficients

In the optical domain, Bogoliubov coefficients serve as a bridge that relates the input and output modes by means of a transformation process that is defined as an isomorphism of either a canonical commutation or anti-commutation relation algebra. The creation and annihilation operators of the input and output modes are related to each other using the Bogoliubov coefficients in the following manner;

$$\hat{a}_{output} = \alpha\hat{a}_{input} + \beta\hat{a}^{\dagger}_{input}$$

$$\hat{a}^{\dagger}_{output} = \alpha^{*}\hat{a}^{\dagger}_{input} + \beta^{*}\hat{a}_{input}$$

As the Bogoliubov transformation is known to be canonical, the commutator of the creation and annihilation operators of the output modes will be;

$$[\hat{a}_{output}, \hat{a}^{\dagger}_{output}] = [\alpha\hat{a}_{input} + \beta\hat{a}^{\dagger}_{input}, \alpha^{*}\hat{a}^{\dagger}_{input} + \beta^{*}\hat{a}_{input}] = (|\alpha|^2 - |\beta|^2)[\hat{a}_{input}, \hat{a}^{\dagger}_{input}]$$

$$\Rightarrow |\alpha|^2 - |\beta|^2 = 1$$

Since $\cosh^2 x - \sinh^2 x = 1$, the Bugoliubov coefficients can easily be expressed as;

$$\alpha = e^{i\theta_1}\cosh r \;\&\; \beta = e^{i\theta_2}\sinh r$$

where, θ_1 & θ_2 correspond to orthogonal symplectic transformations (rotations) and the squeezing factor r corresponds to diagonal transformations

A.1.2 Shift dependent modification to transfer matrix

The additional propagation matrix that results from the shift of the ith interface by a$_i$ units is given by;

$$P(a_i) = \begin{bmatrix} e^{ika_i} & 0 \\ 0 & e^{-ika_i} \end{bmatrix}$$

The modified transfer matrix for the ith interface will therefore be given by;

$$\tilde{M_i} = P(a_i)M_iP^{\dagger}a_i$$

$$\Rightarrow \tilde{M_i} = \begin{bmatrix} e^{ika_i} & 0 \\ 0 & e^{-ika_i} \end{bmatrix} M_i \begin{bmatrix} e^{-ika_i} & 0 \\ 0 & e^{ika_i} \end{bmatrix} = \begin{bmatrix} \alpha & \beta e^{i2ka_i} \\ \beta^{*}e^{-i2ka_i} & \alpha^{*} \end{bmatrix}$$

A.2 Deep dive into two-interface systems

A.2.1 Upper and lower bounds on α_{12} and β_{12}

α_{12} and β_{12} are defined as;

$$\alpha_{12} = \alpha_1\alpha_2 + \beta_1\beta_2^*$$

$$\beta_{12} = \alpha_1\beta_2 + \beta_1\alpha_2^*$$

$$\text{where, } |\alpha_i| = \cosh\theta_i \ \& \ |\beta_i| = \sinh\theta_i$$

The upper bounds corresponding to the two coefficients are obtained from the inequality relations associated with modulus functions. That is;

$$|\alpha_{12}| \leq |\alpha_1||\alpha_2| + |\beta_1||\beta 2| \ \& \ |\beta_{12}| \leq |\alpha_1||\beta_2| + |\beta_1||\alpha_2|$$

In the parametric form;

$$|\alpha_{12}| \leq \cosh\{\cosh^{-1}|\alpha_1| + \cosh^{-1}|\alpha_2|\} \ \& \ |\beta_{12}| \leq \sinh\{\sinh^{-1}|\beta_1| + \sinh^{-1}|\beta_2|\}$$

The upper bound of α_{12} is defined in a similar fashion as;

$$\underbrace{|\alpha_{12}| \geq \cosh\{\cosh^{-1}|\alpha_1| - \cosh^{-1}|\alpha_2|\}}_{\text{The even cosh function accounts for the relative sizes of the } \cosh^{-1}|\alpha_i| \text{ terms}}$$

The derivation of the upper limit of β_{12} is, however, not that straightforward. Since;

$$\beta_{12} = |\alpha_1||\beta_2| + |\beta_1||\alpha_2^*| = |\alpha_1||\beta_2|e^{i(\phi_{\alpha_1}+\phi_{\beta_2})} + |\beta_1||\alpha_2|e^{i(\phi_{-1}-\phi_{\beta_2})}$$

$$\Rightarrow |\beta_{12}| = ||\alpha_1||\beta_2| + |\beta_2||\alpha_1|e^{i(\phi_{\beta_1}-i(\phi_{\beta_2}-i(\phi_{\alpha_1}-i(\phi_{\alpha_2})}$$

As $|\alpha_1||\beta_2|$ and $|\beta_1||\alpha_2|$ are real and positive, $|\beta_{12}|$ will be minimized only when destructive interference is maximum. That is;

$$e^{i(\phi_{\beta_1}-\phi_{\beta_2}-\phi_{\alpha_1}-\phi_{\alpha_1})} = -1$$

Therefore;

$$|\beta_{12} \geq ||\alpha_1||\beta_2| - ||\beta_1||\alpha_2||$$

This gives us;

$$|\beta_{12}| \geq \sinh|\sinh^{-1}|\beta_1| - sinh^{-1}|\beta_2|$$

A.2.2 Hyperbolic identities

Elementary hyperbolic identities are as follows;

$$\begin{aligned}\sinh(\sinh^{-1}A + \sinh^{-1}B) &= \sinh(\sinh^{-1}A)\cosh(sinh^{-1}B) + \cosh(\sinh^{-1}A)\sinh(sinh^{-1}B)\\ &= A\sqrt{1+B^2} + B\sqrt{1+A^2}\end{aligned}$$

$$\begin{aligned}\cosh(\sinh^{-1}A+\sinh^{-1}B) &= \cosh(\sinh^{-1}A)\cosh(sinh^{-1}B)+\sinh(\sinh^{-1}A)\sinh(sinh^{-1}B)\\ &= \sqrt{1+A^2}\sqrt{1+B^2}+AB\end{aligned}$$

$$\begin{aligned}\cosh(\cosh^{-1}A+\cosh^{-1}B) &= \cosh(\cosh^{-1}A)\cosh(cosh^{-1}B)+\sinh(\cosh^{-1}A)\sinh(cosh^{-1}B)\\ &= \sqrt{A^2-1}\sqrt{B^2-1}+AB\end{aligned}$$

$$\tanh(\tanh^{-1}A+\tanh^{-1}B) = \frac{\tanh(\tanh^{-1}A)+\tanh(tanh^{-1}B)}{1+\tanh(\tanh^{-1}A)\tanh(\tanh^{-1}B)} = \frac{A+B}{1+AB}$$

$$\text{sech}(\text{sech}{-}1A+\text{sech}B) = \frac{1}{\cosh(\cosh^{-1}(1/A)+\cosh^{-1}(1/B))} = \frac{AB}{1+\sqrt{1-A^2}\sqrt{1-B^2}}$$

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